

The rank hypotheses in arXiv:1202.1099 are not inherent

Literature-implied-answer status note

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Status. `literature_implied_answer` (full exact question). The implication is identified here from a separate later paper by the same authors; that paper does not cite or explicitly name the 2012 source question.

Original question

The source is D. Alpay and I. Lewkowicz, *State space realization of even generalized positive and odd rational function. Applications to static output feedback*, arXiv:1202.1099 [1].

For a realization with state matrix A , input matrix B , output matrix C , and $A_{cl} = A + BKC$, Lemma 4.1 (source PDF pages 9–10) and Proposition 6.1 (source PDF pages 13–15) establish converse minimality criteria under either

$$\text{rank } B = \text{rank } C \quad \text{or} \quad \min(\text{rank } B, \text{rank } C) \geq r,$$

where r is the number of Jordan blocks of A . Immediately after the two proofs, on source PDF pages 10 and 15, respectively, the authors state that they do not know whether these conditions are inherent or artifacts of their technique.

The two precise converse questions are:

1. Does minimality alone imply that some constant K satisfies

$$\text{spect}(A) \cap \text{spect}(A + BKC) = \emptyset?$$

2. In the square realization-matrix setting, does minimality alone imply

$$\left(\bigcap_{D \in \mathbb{C}^{p \times p}} \text{spect} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \right) \cap \text{spect}(A) = \emptyset?$$

Later theorem

The later paper D. Alpay and I. Lewkowicz, *Minimal state space realization, static output feedback and matrix completion of families of systems*, arXiv:1305.1756v2 [2], gives exactly the unconditional answer in Theorem 1.2 on supporting PDF pages 3–4.

Theorem 1.2 states the equivalence of:

- (i) the realization is minimal;
- (ii) there exists one constant $K \in \mathbb{C}^{m \times p}$ such that

$$\text{spect}(A) \cap \text{spect}(A + BKC) = \emptyset;$$

(iv) in the square case (or after the paper’s minimal square reduction),

$$\left(\bigcap_{D \in \mathbb{C}^{p \times p}} \text{spect} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \right) \cap \text{spect}(A) = \emptyset.$$

No equality or lower bound involving rank B , rank C , or the number of Jordan blocks appears in these equivalences. Therefore the source hypotheses are not inherent: minimality alone suffices for both formal converse statements.

Provenance and scope

This is an agent-identified implication, not a new proof and not an explicit claim by the supporting paper that it resolves arXiv:1202.1099. The match is nevertheless exact: the authors are the same, the feedback matrix is the same unstructured constant K , and Theorem 1.2(ii) is the rank-free version of source Proposition 6.1. Item (iv) is the rank-free version of the formal converse in source Lemma 4.1. The supporting paper also handles rectangular systems by its minimal square reduction, a modest extension beyond the source lemma’s square setting.

The identification does not address structured or decentralized feedback, nor arbitrary pole placement by static output feedback. Those are different problems from avoiding the finite set $\text{spect}(A)$.

Search evidence. The run’s four lightweight indexes contained no prior entry for either arXiv id. Primary arXiv searches for the exact source title, “static output feedback fixed modes”, and “minimal realization $A + BKC$ ” found arXiv:1305.1756. Its Theorem 1.2 was then checked in the source TeX and PDF.

References

- [1] D. Alpay and I. Lewkowicz, *State space realization of even generalized positive and odd rational function. Applications to static output feedback*, arXiv:1202.1099 (2012).
- [2] D. Alpay and I. Lewkowicz, *Minimal state space realization, static output feedback and matrix completion of families of systems*, arXiv:1305.1756v2 (2013).